

Xingjian Bai

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Education

Ph.D. in EECS, Massachusetts Institute of Technology

Sep 2024 – Present

- MongoDB PhD Fellowship.
- Robert M. (1941) and Jacqueline M. Fano Fellowship.
- Research focus: generative models in computer vision.
- Expected graduation: 2028.

Master of Mathematics and Computer Science, University of Oxford

Oct 2023 – Jul 2024

- Received the **Departmental Prize for the Best Computer Science Project**.

BA in Mathematics and Computer Science, University of Oxford

Oct 2020 – Jul 2023

- Graduated with Distinction and awarded the **Gibbs Prize** for the best performance in Computer Science examinations.

Research Experience

Adobe Research

Research Scientist Intern

Jun 2026 – Present

Supervisor: Richard Zhang

Adobe Research

Research Scientist Intern

Jun 2025 – Mar 2026

Supervisors: Xun Huang, Zongze Wu

Investigated causal structure in video diffusion models and developed methods that decouple temporal reasoning from denoising for more efficient video generation.

Visual Geometry Group (VGG), Oxford

Student Researcher

Oct 2023 – Apr 2024

Supervisors: Prof. Christian Rupprecht, Luke Melas-Kyriazi

Designed fixed-point diffusion models that decouple denoising depth from sampling steps, enabling flexible allocation of computation across the diffusion process.

Stanford Vision & Learning Lab (SVL)

Visiting Research Intern

Jul 2023 – Sep 2023

Supervisor: Prof. Jiajun Wu

Studied neurosymbolic controllers for diffusion models to improve compositionality.

Algorithms and Complexity Theory Group, Oxford

Undergraduate Student Researcher

Mar 2023 – Aug 2023

Supervisor: Prof. Christian Coester

Developed sorting algorithms that exploit imperfect machine-learning predictions to improve upon the classical $O(n \log n)$ running time under mild assumptions.

Mathematics Institute, Oxford

Summer Research Intern

Jul 2022 – Apr 2023

Supervisor: Prof. Jan Oblój

Developed adversarial attack and robustness-analysis methods for neural networks using sensitivity results from distributionally robust optimization (DRO).

AI Safety Research Lab, Oxford

Student Researcher

Nov 2022 – Mar 2023

Mentor: Joar Skalse

Studied reward hacking caused by over-optimization in reinforcement learning, developed a geometric explanation, and proposed an early-stopping method to mitigate it.

Selected Publications

*Equal contribution.

Sizhe Lester Li*, Evan Kim*, **Xingjian Bai***, Tong Zhao, Tao Pang, Max Simchowitz, Vincent Sitzmann.

“Turning Video Models into Generalist Robot Policies.”

arXiv preprint, 2026. [arXiv]

Shivam Duggal*, **Xingjian Bai***, Zongze Wu, Richard Zhang, Eli Shechtman, Antonio Torralba, Phillip Isola, William T. Freeman.

“End-to-End Training for Unified Tokenization and Latent Denoising.”

arXiv preprint, 2026. [arXiv]

Selected Publications (continued)

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- Xingjian Bai**, Guande He, Zhengqi Li, Eli Shechtman, Xun Huang, Zongze Wu.
“Causality in Video Diffusers is Separable from Denoising.”
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. [arXiv]
- Zhengyang Geng, Mingyang Deng, **Xingjian Bai**, J. Zico Kolter, Kaiming He.
“Mean Flows for One-step Generative Modeling.”
Advances in Neural Information Processing Systems (NeurIPS), 2025. **Oral**. [arXiv]
- Xingjian Bai**, Christian Coester, Romain Cosson.
“Unweighted Layered Graph Traversal: Passing a Crown via Entropy Maximization.”
ACM-SIAM Symposium on Discrete Algorithms (SODA), 2025. [arXiv]
- Xingjian Bai**^{*}, Luke Melas-Kyriazi^{*}.
“Fixed Point Diffusion Models.”
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. [arXiv]
- Xingjian Bai**, Christian Coester.
“Sorting with Predictions.”
Advances in Neural Information Processing Systems (NeurIPS), 2023. [arXiv]
- Xingjian Bai**, Guangyi He, Yifan Jiang, Jan Oblój.
“Wasserstein Distributional Robustness of Neural Networks.”
Advances in Neural Information Processing Systems (NeurIPS), 2023. [arXiv]

Awards & Honors

World Finalist & Regional Gold Medalist <i>International Collegiate Programming Contest (ICPC)</i>	2023
Outstanding Winner & AMS Best Paper Award (1 of 10,053 teams) <i>37th Mathematical Contest in Modeling (MCM)</i>	2021
“Hack the Hackers’ Hack” Award (Best of 66 Teams) <i>Oxford Hackathon</i>	2020
Gold Medalist (First Place) <i>Canadian Computing Olympiad (CCO), national team selection camp</i>	2018
Silver Medalist <i>Chinese National Olympiad in Informatics (NOI)</i>	2018

Other Experience

Conference Reviewer <i>NeurIPS, ICLR, ICML, CVPR, AISTATS</i>	Present
Class Demonstrator, Computer Vision <i>Department of Computer Science, University of Oxford</i>	2024
Student Ambassador <i>Mathematics Institute, University of Oxford</i> Supported outreach events and taught algorithms to students from under-resourced regions.	2023

Skills & Interests

Programming: Proficient in C++ and Python; experience with Julia, Java, Scala, and Haskell.
ML & Tools: PyTorch, JAX/Flax, Linux, Git, basic CUDA.
Interests: Marathon running, tennis, table tennis, and Go.